



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 - 2022)

JEE(Advanced)
PRE-SCORE
17-11-2021

JEE(Main + Advanced) : ENTHUSIAST COURSE (PHASE : I(A), TTAS & TEAS)

PAPER-2

Time : 3 Hours

Maximum Marks : 198

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **24** pages and that all the **18** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

QUESTION PAPER FORMAT :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

Please see the last page of this booklet for rest of the instructions

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba = 137, Ce = 140

- **Boltzmann constant** $k = 1.38 \times 10^{-23} \text{ JK}^{-1}$
- **Coulomb's law constant** $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$
- **Universal gravitational constant** $G = 6.67259 \times 10^{-11} \text{ N-m}^2 \text{ kg}^{-2}$
- **Speed of light in vacuum** $c = 3 \times 10^8 \text{ ms}^{-1}$
- **Stefan-Boltzmann constant** $\sigma = 5.67 \times 10^{-8} \text{ Wm}^{-2}\text{-K}^{-4}$
- **Wien's displacement law constant** $b = 2.89 \times 10^{-3} \text{ m-K}$
- **Permeability of vacuum** $\mu_0 = 4\pi \times 10^{-7} \text{ NA}^{-2}$
- **Permittivity of vacuum** $\epsilon_0 = \frac{1}{\mu_0 c^2}$
- **Planck constant** $h = 6.63 \times 10^{-34} \text{ J-s}$

Space for Rough Work

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

BEWARE OF NEGATIVE MARKING

PART-1 : PHYSICS

SECTION-I : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

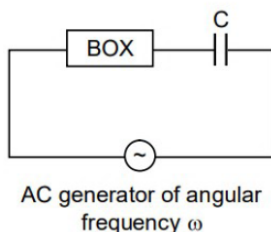
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

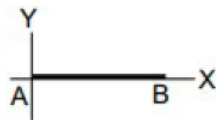
1. A box has a large electric circuit inside it. When it was connected to an AC generator, it was found that it was putting a lot of load on the generator and the power factor of the box was $\frac{1}{\sqrt{2}}$. Now a capacitor of capacitance C was connected in series with the box and the power factor of the circuit becomes equal to 1. Then



- (A) The impedance of the box must be resistive and capacitive.
- (B) The impedance of the box must be resistive and inductive.
- (C) The impedance of the box is $\frac{\sqrt{2}}{\omega C}$
- (D) The impedance of the box is $\frac{1}{\sqrt{2} \omega C}$

Space for Rough Work

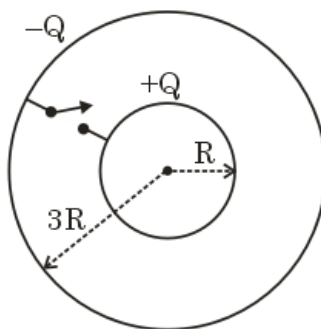
2. A wire of mass M and length L is placed along x -axis with end A at origin. The linear mass density λ (of the rod) varies as $\lambda = \lambda_0 x$, where x is distance from end A , λ_0 is a constant.



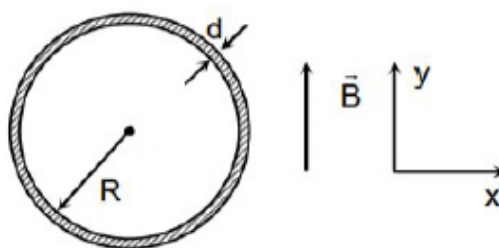
- (A) The value of λ_0 is $\frac{2M}{L^2}$
- (B) Moment of Inertia of the rod about an axis passing through A and parallel to y -axis is $\frac{1}{2}ML^2$
- (C) Moment of inertia of the rod about an axis passing through COM of the rod and parallel to y -axis is $\frac{ML^2}{18}$.
- (D) Moment of inertia of the rod about an axis passing through B and parallel to y -axis is $\frac{1}{6}ML^2$.
3. X-ray from a tube with a target A of atomic number Z shows strong K lines for target A and weak K lines for impurities. The wavelength of K_α lines is λ_z for target A and λ_1 and λ_2 for two impurities. If $\frac{\lambda_z}{\lambda_1} = 4$ and $\frac{\lambda_z}{\lambda_2} = \frac{1}{4}$. Screening constant of K_α lines to be unity. Select the correct statement(s)
- (A) The atomic number of first impurity is $2Z - 1$
- (B) The atomic number of first impurity is $2Z + 1$
- (C) The atomic number of second impurity is $\frac{(Z + 1)}{2}$
- (D) The atomic number of second impurity is $\frac{Z}{2} + 1$

Space for Rough Work

4. Shown in figure are two uniformly charged conducting spherical thin shells. Total electric potential energy (before closing switch) 'U' of the system and heat loss 'H' when switch is closed.



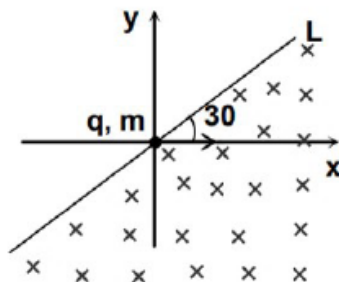
- (A) $H = \frac{kQ^2}{12R}$ (B) $H = \frac{kQ^2}{3R}$ (C) $U = \frac{kQ^2}{2R}$ (D) $U = \frac{kQ^2}{3R}$
5. A thin walled non-magnetic spherical shell with wall thickness d and radius R ($R \gg d$) is placed in a uniform magnetic field $\vec{B} = B_0 \hat{j}$, which slowly changes as $\frac{dB_0}{dt} = \alpha$, where α is a positive constant. Electrical resistivity of the material of the shell is ρ . Then choose the correct option(s).



- (A) Induced magnetic moment of the shell is towards $+\hat{j}$ direction
 (B) Induced magnetic moment of the shell is towards $-\hat{j}$ direction
 (C) Induced magnetic moment of the shell is $\frac{3\pi}{2\rho} dR^4 \alpha$
 (D) Induced magnetic moment of the shell is $\frac{2\pi}{3\rho} dR^4 \alpha$

Space for Rough Work

6. A charged particle of charge q and mass m is entering at origin with a velocity $v\hat{i}$ in a uniform magnetic field $-B\hat{k}$ which is extended upto infinite at right of the line L . Select the correct option(s). Given that $\frac{mv}{qB} = 1$ meter, and $v = 1$ m/s.



- (A) Velocity of the particle makes an angle 60° with x-axis while it is coming out of the magnetic field.
- (B) The co-ordinates of point of exit is $\left(\frac{1}{2}m, \frac{\sqrt{3}}{2}m\right)$.
- (C) The co-ordinates of point of exit is $\left(\frac{\sqrt{3}}{2}m, \frac{1}{2}m\right)$.
- (D) Time spent by the particle in the magnetic field is $\frac{\pi}{3}$ sec.

Space for Rough Work

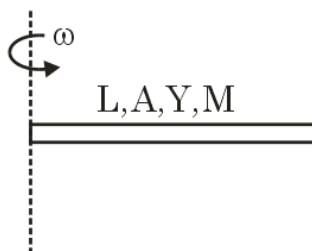
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- This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE**.
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- Answer to each question will be evaluated according to the following marking scheme:

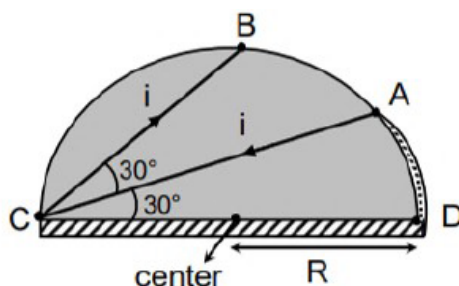
Full Marks : +4 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

1. A rod of mass M , length L , area of cross section A and young modulus of elasticity Y rotating with angular speed about axis passing through one end and perpendicular to it's length then find extension in the length of rod due to rotation. (given : $\omega = \sqrt{\frac{AY}{mL^2}}$)

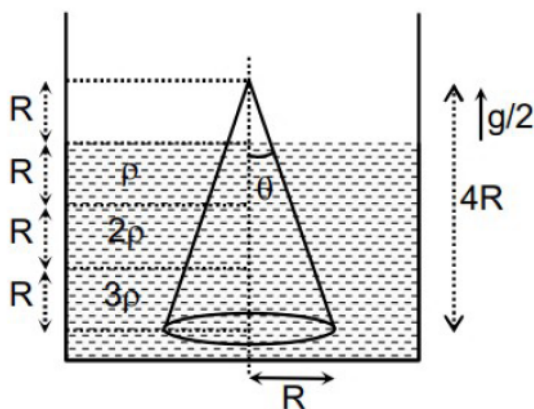


2. A current carrying wire is placed in the grooves of an insulating semicircular disc of radius R as shown in the figure. The current enters at point A and leaves from point B. The net magnetic field at point D due to AC and CB is $B_0 \times 10^{-7} \text{ T}$, find value of B_0 . (Given $i = \sqrt{3} \text{ Amp}$, $R = 4 \text{ m}$)



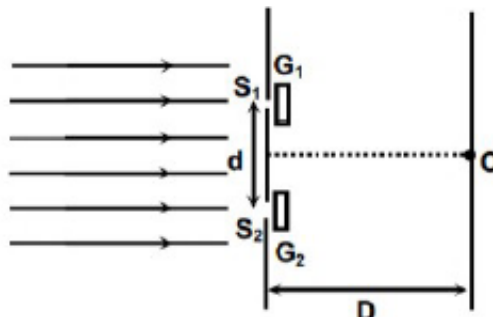
Space for Rough Work

3. A solid cone is kept in three immiscible liquids of density ρ , 2ρ and 3ρ inside a container. The radius of cone is ' R ' and its height is $4R$. Container is accelerated with a constant acceleration $g/2$ in vertically upward direction as shown. If cone is in equilibrium relative to the frame of container then density of the cone is $a\rho$. Find value of a .

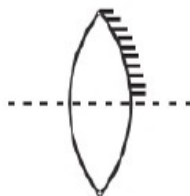


Space for Rough Work

4. A parallel beam of monochromatic light of wavelength $4500 \text{ \AA}$ is incident on YDSE. Let I_0 is the intensity of incident light. Now two glass slabs G_1 and G_2 are kept in front of both slits S_1 and S_2 . The refractive index of G_1 is 1.9 and for G_2 it is 1.6. The thickness of each glass slab is $15 \mu\text{m}$. It is observed that glass G_1 transmits $\frac{1}{16}$ th and G_2 transmits $\frac{1}{25}$ th of incident energy. If $I_0 = 8 \text{ Watt/m}^2$, then find the intensity (in W/m^2) observed at point 'O' on the screen.



5. Half part of the rear surface of a thin equiconvex lens of focal length 40 cm and refractive index 1.5 is silvered. If parallel rays are incident on this lens, then the distance (in cm) between the two images.



6. A plano-convex lens has a thickness of 4 cm. When placed on a horizontal table with curved surface in contact with it, the apparent depth of the bottom most point of the lens is found to be 3 cm. If the lens is inverted such that the plane face is in contact with the table, the apparent depth of the centre of plane face is found to be $\frac{25}{8}$ cm. Find the focal length of lens (in cm)

Space for Rough Work

SECTION-III : (Maximum Marks: 18)

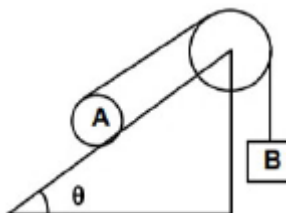
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Negative Marks : -1 In all other cases

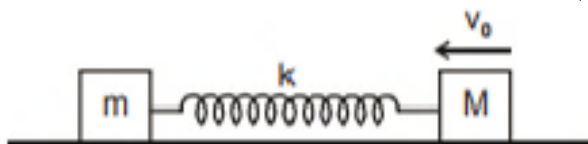
1. In the figure shown, a string is wound over a cylinder A. The other end of the string is attached to block B through a pulley. When the system is released the cylinder rolls down without slipping. The magnitude of vertical component of displacement of A and B in any time interval t is y_A and y_B . Find value of $\frac{10y_A}{y_B}$. (Use $\theta = 37^\circ$)



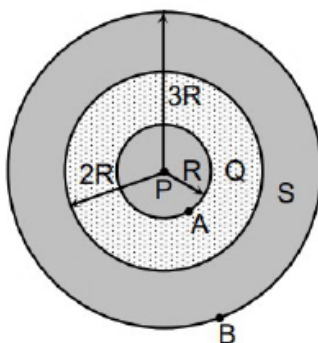
2. The equation of a transverse wave is given by $\psi = 10^{-2} \sin \pi [30t - x\sqrt{3} - y]$ where x , y and ψ are in metre, and t in second. If phase difference between two points A $(2\sqrt{3}\text{m}, 2\text{m})$ and B $(3\sqrt{3}\text{m}, 3\text{m})$ be $n\pi$. Find value of n
3. A cylinder of mass M radius R is resting on a horizontal platform (which is parallel to x - y plane) with its axis fixed along the y -axis and free to rotate about its axis. The platform is given a motion in X -direction given by $X = A \cos \omega t$. There is sufficient friction present in the surface of contact that can prevent the slipping between the cylinder and platform. The maximum torque in N-m acting on the cylinder during its motion is. [Take $M = 4 \text{ kg}$, $R = 1 \text{ m}$, $A = 2 \text{ m}$, $\omega = 1 \text{ rad/s}$]

Space for Rough Work

4. A thin biprism of obtuse angle 178° is placed at a distance $\ell = \frac{60}{\pi}$ cm from a slit. Images are formed then what is the separation (in mm) between the images? Refractive index of material of prism is $\mu = 1.6$.
5. Two blocks of masses m and M are joined with an ideal spring of spring constant k and kept on a rough surface as shown. The spring is initially unstretched and the coefficient of friction between the blocks and the horizontal surface is μ . What should be the maximum speed of the block of mass M such that the smaller block does not move? (Use $M = 1\text{ kg}$, $m = 2\text{ kg}$, $k = 100\text{ N/m}$, $\mu = \frac{1}{\sqrt{8}}$, $g = 10\text{ m/s}^2$)



6. A solid sphere P and two hollow sphere Q and S are concentrically placed as shown in the figure. The radius of P, Q and S are R , $2R$ and $3R$ respectively. The density of P, Q and S are ρ_1 , ρ_2 and ρ_3 respectively. It is given that the density of middle sphere is the arithmetic mean of the densities of other two spheres. If the gravitational fields at points A and B are equal then find the ratio of $\frac{\rho_3}{\rho_1}$ and fill the value of $10 \frac{\rho_3}{\rho_1}$.



Space for Rough Work

PART-2 : CHEMISTRY
SECTION-I : (Maximum Marks: 24)

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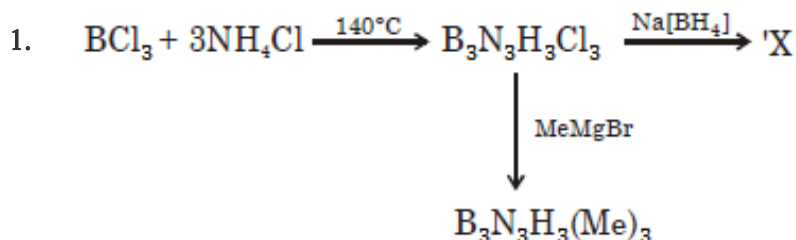
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

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- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.



Which of the following statements are CORRECT :-

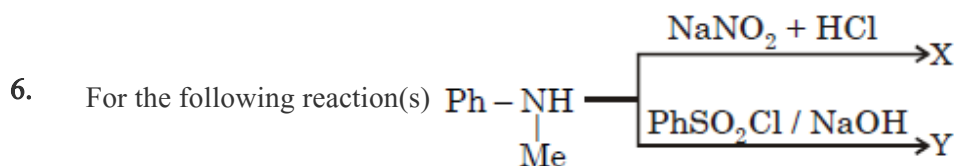
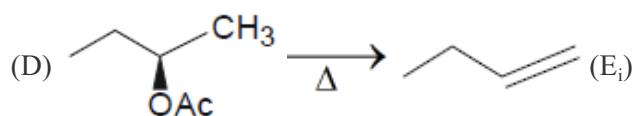
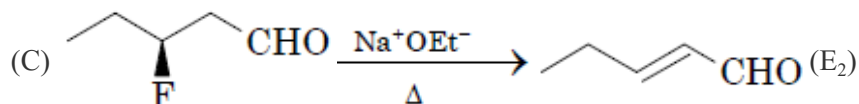
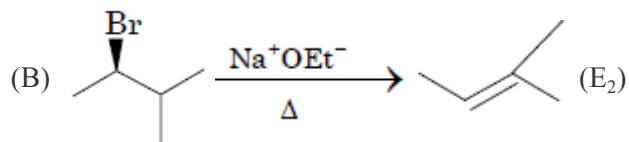
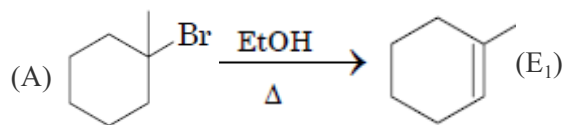
- (A) 'X' is known as inorganic graphite
- (B) 'X' on hydrolysis along with heating gives NH_3 , H_2 and B(OH)_3
- (C) 'X' is known as inorganic benzene
- (D) 'X' has $2p\pi-2p\pi$ back bonding

Space for Rough Work

2. Which of the following pairs have approximately the same atomic radii :
- (A) Zr & Hf (B) Al, Mg (C) Nb, Ta (D) Na, Ne
3. In a compound oxide ions arranged in CCP pattern. One-eight of the tetrahedral holes occupied by one type of metal ion and one half of the octahedral hole occupied by another type of metal ion. If metal ions are A^{+2} & B^{+3} , then select the correct statements -
- (A) A^{+2} occupy tetrahedral holes (B) B^{+3} occupy tetrahedral holes
(C) Formula of compound is A_2BO_4 (D) Formula of compound is AB_2O_4
4. Select the correct statement(s)
- (A) Suspensions may show Tyndall effect.
(B) The reactions taking place in zeolites depend upon the size and shape of reactant and product molecules as well as upon the nature of pores and cavities of zeolites.
(C) Powdered substances are better adsorbents than their crystalline forms.
(D) Tyndall effect may appear if excess $AgNO_3$ solution is added into aqueous solution of KI.
-

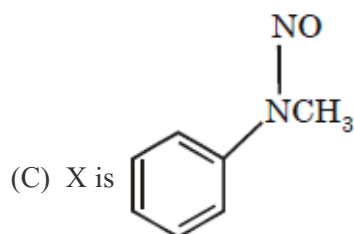
Space for Rough Work

5. Which among the following reactions are correctly matched with their major product and mechanism of the reaction :



Correct options for possible products X & Y is/are ?

- (A) X is oily liquid
(B) Y is water insoluble and does not react with base



- (D) In Y lone pair of nitrogen is delocalised

Space for Rough Work

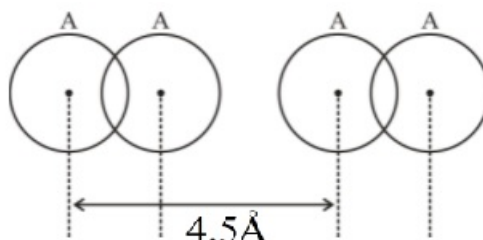
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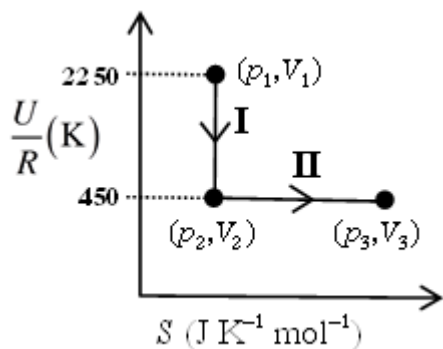
1. Out of given chemicals, no. of chemicals which is/are related to Solvay process :
 NH_3 , KCl , CO_2 , SO_2 , Ca(OH)_2 , NaCl , H_2SO_4 , NaNO_3 :
2. Covalent radius of A = $1\text{\AA}$
 Vander Walls's radius of A (in ($\text{\AA}$)) will be....



3. 0.8 gram-atoms of an α -emitting radionuclide ${}^A_Z\text{X}$ ($t_{1/2} = 2$ hours.) were placed in a sealed container. If 15.68 litres of He(g) was collected in container kept in 1atm and 273K conditions in t hours. Find value of 't' ? (Daughter nucleus formed by α -decay is non-radioactive)

Space for Rough Work

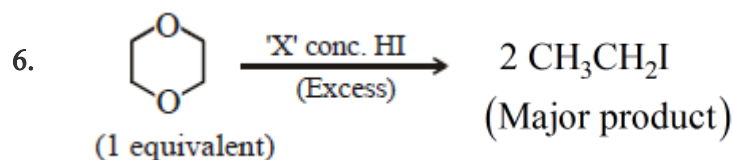
4. One mole of an ideal gas at 900 K, undergoes two reversible processes, I followed by II, as shown below. If the work done by the gas in the two processes are same, the value of $\ln \frac{V_3}{V_2}$ is ____.



(U : internal energy, S : entropy, p : pressure, V : volume, R : gas constant)

(Given : molar heat capacity at constant volume, $C_{V,m}$ of the gas is $\frac{5}{2} R$)

5. Number of chiral center's in α -D-glucopyranose?



Values 'X' for complete reaction

Space for Rough Work

SECTION-III : (Maximum Marks: 18)

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1. Find the number of pair(s) having wrong order of indicated property

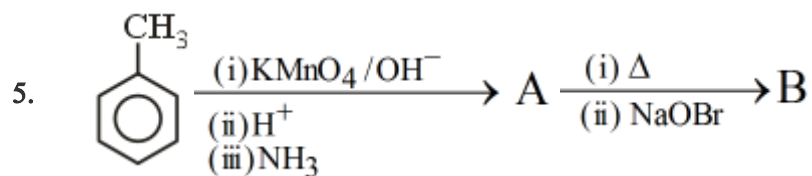
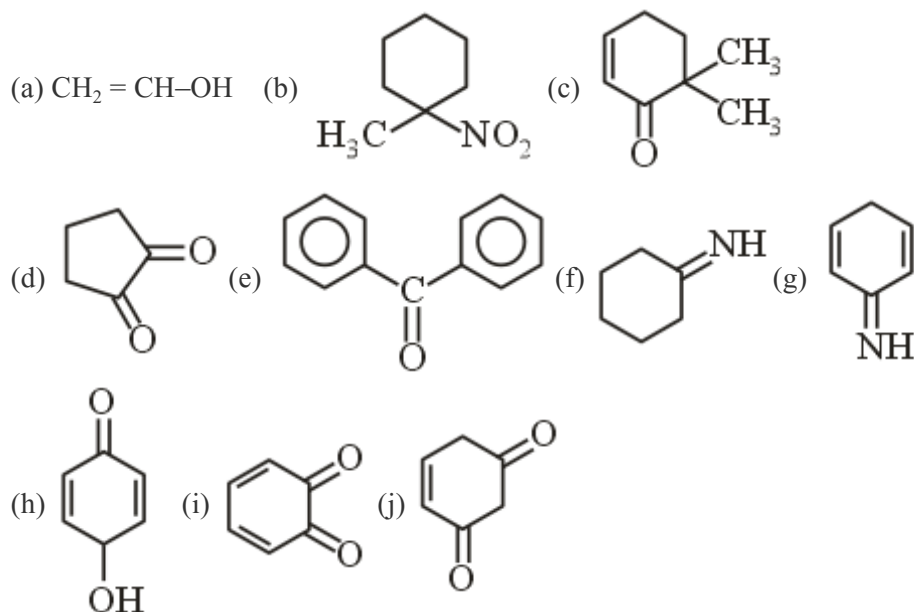
(i) $\text{SF}_6 < \text{TeF}_6$ (tendency of hydrolysis)	(ii) $\text{H}_2 > \text{He}$ (boiling point)
(iii) $\text{BF}_3 < \text{BCl}_3$ (bond angle)	(iv) $\text{CaCO}_3 > \text{BaCO}_3$ (thermal stability)
(v) $\text{PbO} < \text{PbO}_2$ (oxidizing property)	

2. Equivalent conductance of saturated solution of BaSO_4 is $400 \text{ ohm}^{-1}\text{cm}^2\text{eq}^{-1}$ and specific conductance is $8 \times 10^{-5} \text{ ohm}^{-1}\text{cm}^{-1}$. K_{sp} of BaSO_4 is 1×10^{-x} , then find x

3. $\text{KI}_{(\text{aq.})}$ and $\text{KIO}_{3(\text{aq.})}$ mixed together in presence of HCl to form I_2 and KCl and liberated I_2 required 100 mL of 0.1 M $\text{Na}_2\text{S}_2\text{O}_3$ solution for complete reduction of I_2 .
 If x = millimoles of KI used in the reaction.
 and y = millimoles of HCl used in the reaction.
 then value of $(3x - 2y)$ is :

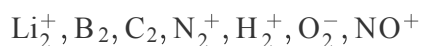
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4. How many of the following compounds can exhibit tautomerism?



Total number of sp^2 hybridised carbon in A if x and total number of hydrogen present in B if y then find out x/y ?

6. Among the following, total number of species is/are paramagnetic, as well as having fractional bond order :-



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PART-3 : MATHEMATICS
SECTION-I : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

1. If $f(x) = x^5 \text{sgn}(\text{sgn}(x))$, where $\text{sgn}(\cdot)$ denotes signum function, then which is/are correct

- (A) $f(x)$ is discontinuous at $x = 0$
- (B) $f(x)$ is continuous but not differentiable at $x = 0$
- (C) $f'(0^+) = 0$
- (D) $f(x)$ is continuous as well as differentiable at $x = 0$

2. Which of the following functions are identical functions -

- (A) $f(x) = \tan^{-1}x + \tan^{-1}(-x)$ & $g(x) = 2\tan^{-1}x$
- (B) $f(x) = \cos^{-1}x + \cos^{-1}(-x)$ & $g(x) = 2\cos^{-1}x$
- (C) $f(x) = \cos x + \cos|x|$ & $g(x) = 2 \cos x$
- (D) $f(x) = \cos^{-1}x + |\cos^{-1}x|$ & $g(x) = 2 \cos^{-1}x$

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3. The possible solution of $(\cos^2 x - 4\sin x + 6)(1 - \sin x) = \cos^2 x$, is/are
 (A) $\frac{\pi}{2}$ (B) $\frac{3\pi}{2}$ (C) $\frac{5\pi}{2}$ (D) $\frac{7\pi}{2}$
4. If $\lim_{x \rightarrow 0} (a \sin bx^2 - 4\cos^2 x + 5) \frac{4}{5x^2} = e^2$ where $a, b \in \mathbb{R}$ then the value of a & b can be -
 (A) $-3, \frac{1}{2}$ (B) $3, -\frac{1}{2}$ (C) $6, -\frac{1}{4}$ (D) $1, -\frac{3}{2}$
5. Let $f : [-1, 1] \rightarrow [-1, 1]$, $f(x) = \sin(2\sin^{-1}x)$ & let $g(x) = \frac{f(x)}{\sqrt{1-x^2}}$, then -
 (A) $f(x)$ is one-one function (B) $f(x)$ is odd function
 (C) $g(x)$ is continuous in $(-1, 1)$ (D) $f(x)$ is continuous for all $x \in [-1, 1]$
6. Let $y = y(x)$, $y(1) = 1$ and $y(e) = e^2$. Consider $J = \int \frac{x+y}{xy} dy$, $I = \int \frac{x+y}{x^2} dx$, $J - I = g(x)$ and $g(1) = 1$, then the value of $g(e)$ is -
 (A) $3e + 1$ (B) $e + 1$ (C) $e^2 - e + 2$ (D) $e^2 + e - 2$

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SECTION-II : (Maximum Marks: 24)

- This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

1. If $D = \begin{bmatrix} 0 & a\alpha^2 & a\beta^2 \\ b\alpha + c & 0 & a\gamma^2 \\ b\beta + c & (b\gamma + c) & 0 \end{bmatrix}$ is a skew symmetric matrix (where α, β, γ are distinct) & the

value of $\begin{vmatrix} (a+1)^2 & (1-a) & (2-c) \\ (3+c) & (b+2)^2 & (b+1)^2 \\ (3-b)^2 & b^2 & (c+3) \end{vmatrix}$ is λ then $|10\lambda|$ is equal to

2. Let $f(x)$ be a derivable function satisfying $\int_0^x f(t)dt = k + f^2(x)$ where $f(1) = \frac{7}{2}$ then $|k|$ is equal to
3. If the locus of the point of intersection of two tangents to a parabola $y^2 = 4x$ at the points whose ordinates have the constant ratio 4 : 1 can be expressed as $y^2 = \frac{p}{q}x$ (where p and q are integers in lowest form), then $(p + q)$ is
4. Let $f(x) = ax^2 + bx + c$, $a, b, c \in I$ & $f(1) = 0$, $50 < f(7) < 60$ & $70 < f(8) < 80$, then $f(-1)$ is equal to -
5. If $f : [2, \infty) \rightarrow R$; $f(x)$ is a function which is differentiable as well as increasing & $f(2) = 2$; then the number of solution of equation $f(f(x)) = \frac{4}{(x^2 - 4x + 6)}$ is
6. Let $|z| = |z - 3| = |z - 4i|$, then $|2z|$ is equal to

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SECTION-III : (Maximum Marks: 18)

- This section contains **SIX (06)** questions.
- The answer to each question is a **SINGLE DIGIT INTEGER** ranging from 0 to 9, both inclusive.
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +3 If only the correct answer is given.

Zero Marks : 0 If no answer is given.

Negative Marks : -1 In all other cases

1. If $\int_{-5}^{-2} \left(\frac{x^2 - x}{x^3 - 3x + 1} \right)^2 dx + \int_{1/6}^{1/3} \left(\frac{x^2 - x}{x^3 - 3x + 1} \right)^2 dx + \int_{6/5}^{3/2} \left(\frac{x^2 - x}{x^3 - 3x + 1} \right)^2 dx = \frac{p}{q}$, where p & q are co-primes, then $\frac{6q - p}{6}$ is equal to
2. A triangle is formed by the points A(0, 0), B(3, 0) and C(3, 4). A and C are foci of ellipse and B lies on the ellipse. If area of ellipse is $\frac{7\pi}{2} \sqrt{P}$ ($P \in \mathbb{N}$), then the value of P is
3. A box contains 28 identical balls of which 21 are red and 7 are black. The balls are drawn at random from the box one at a time with replacement. If the Probability that red ball is drawn for the 5th time on the 13th draw is $\frac{55 \cdot 3^5 k}{4^{13}}$, then k is
4. The maximum volume of the parallelopiped whose sides are given by $4\hat{i} + 2\cos\theta\hat{j} + 2\hat{k}$, $2\hat{i} + \hat{j} + \cos\theta\hat{k}$ and $2\hat{i} + \cos\theta\hat{j} + \sin\theta\hat{k}$ is V, then the value of $\left\lceil \frac{V}{2} \right\rceil$ (where $\lceil . \rceil$ denotes greatest integer function and $\theta \in \mathbb{R}$)
5. If a, b, c be last three digits of the number $(6! + 1)^{6!}$, then $(a + b + c)$ is
6. If $\frac{dy}{dx} = \frac{e^x(\sin x + \cos x)}{(\tan y + y \sec^2 y)}$; if the curve passes through origin & (x_1, π) , where $6 < x_1 < 7$, then $\frac{x_1}{\pi}$ is equal to

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NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions
and shall abide by them.

Signature of the Candidate

I have verified the identity, name and Form
number of the candidate, and that question
paper and ORS codes are the same.

Signature of the Invigilator

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